CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

# Chapter 13: Statistics

Step-by-step solutions to the textbook exercises: 22 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

## 1 Exercise 13.1

**Q1** The numbers of plants in 20 houses are given below. Find the mean number of plants per house. Which method did you use, and why?

| Number of plants | 0 – 2 | 2 – 4 | 4 – 6 | 6 – 8 | 8 – 10 | 10 – 12 | 12 – 14 |
|---|---|---|---|---|---|---|---|
| Number of houses | 1 | 2 | 1 | 5 | 6 | 2 | 3 |

SOLUTION

The class marks and frequencies are small numbers, so the direct method is the easiest.

| Class mark $x_i$ | 1 | 3 | 5 | 7 | 9 | 11 | 13 | Total |
|---|---|---|---|---|---|---|---|---|
| $f_i$ | 1 | 2 | 1 | 5 | 6 | 2 | 3 | 20 |
| $f_i x_i$ | 1 | 6 | 5 | 35 | 54 | 22 | 39 | 162 |

Mean $= \dfrac{\sum f_i x_i}{\sum f_i} = \dfrac{162}{20} = 8.1.$

**Answer: 8.1 plants per house, by the direct method**

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**Q2** The daily wages of 50 workers of a factory are given below. Find the mean daily wage by a suitable method.

| Daily wages (₹) | 500 – 520 | 520 – 540 | 540 – 560 | 560 – 580 | 580 – 600 |
|---|---|---|---|---|---|
| Number of workers | 12 | 14 | 8 | 6 | 10 |

SOLUTION

Use the step-deviation method with $a = 550$, $h = 20$ and $u_i = \dfrac{x_i - a}{h}$.

| $x_i$ | 510 | 530 | 550 | 570 | 590 | Total |
|---|---|---|---|---|---|---|
| $f_i$ | 12 | 14 | 8 | 6 | 10 | 50 |
| $u_i$ | $-2$ | $-1$ | 0 | 1 | 2 | |
| $f_i u_i$ | $-24$ | $-14$ | 0 | 6 | 20 | $-12$ |

$$\text{Mean} = a + \frac{\sum f_i u_i}{\sum f_i} \times h = 550 + \frac{-12}{50} \times 20 = 550 - 4.8 = 545.2.$$

**Answer: ₹545.20**

**Q3** The daily pocket allowance of the children of a locality is given below. The mean is ₹18. Find the missing frequency $f$.

| Allowance (₹) | 11 – 13 | 13 – 15 | 15 – 17 | 17 – 19 | 19 – 21 | 21 – 23 | 23 – 25 |
|---|---|---|---|---|---|---|---|
| Number of children | 7 | 6 | 9 | 13 | $f$ | 5 | 4 |

SOLUTION

The class marks are 12, 14, 16, 18, 20, 22, 24.

$\sum f_i = 44 + f$ and $\sum f_i x_i = 84 + 84 + 144 + 234 + 20f + 110 + 96 = 752 + 20f$.

$\dfrac{752 + 20f}{44 + f} = 18$ gives $752 + 20f = 792 + 18f$, so $2f = 40$.

**Answer:** $f = 20$

**Q4** The heartbeats per minute of 30 women are summarised below. Find the mean by a suitable method.

| Heartbeats per minute | 65 – 68 | 68 – 71 | 71 – 74 | 74 – 77 | 77 – 80 | 80 – 83 | 83 – 86 |
|---|---|---|---|---|---|---|---|
| Number of women | 2 | 4 | 3 | 8 | 7 | 4 | 2 |

SOLUTION

Use the step-deviation method with $a = 75.5$ and $h = 3$.

| $x_i$ | 66.5 | 69.5 | 72.5 | 75.5 | 78.5 | 81.5 | 84.5 | Total |
|---|---|---|---|---|---|---|---|---|
| $f_i$ | 2 | 4 | 3 | 8 | 7 | 4 | 2 | 30 |
| $u_i$ | $-3$ | $-2$ | $-1$ | 0 | 1 | 2 | 3 | |
| $f_iu_i$ | $-6$ | $-8$ | $-3$ | 0 | 7 | 8 | 6 | 4 |

Mean $= 75.5 + \dfrac{4}{30} \times 3 = 75.5 + 0.4 = 75.9$.

**Answer: 75.9 heartbeats per minute**

**Q5** Boxes of mangoes hold different numbers of mangoes, as shown below. Find the mean number of mangoes in a box. Which method did you choose?

| Number of mangoes | 50 – 52 | 53 – 55 | 56 – 58 | 59 – 61 | 62 – 64 |
|---|---|---|---|---|---|
| Number of boxes | 15 | 110 | 135 | 115 | 25 |

SOLUTION

The class marks are 51, 54, 57, 60, 63. (They are the same whether or not the classes are made continuous.) Use the step-deviation method with $a = 57$ and $h = 3$.

| $x_i$ | 51 | 54 | 57 | 60 | 63 | Total |
|---|---|---|---|---|---|---|
| $f_i$ | 15 | 110 | 135 | 115 | 25 | 400 |
| $u_i$ | $-2$ | $-1$ | 0 | 1 | 2 | |
| $f_iu_i$ | $-30$ | $-110$ | 0 | 115 | 50 | 25 |

Mean $= 57 + \dfrac{25}{400} \times 3 = 57 + 0.1875 \approx 57.19$.

**Answer: about 57.19 mangoes, by the step-deviation method**

**Q6** The daily expenditure on food of 25 households is given below. Find the mean by a suitable method.

| Daily expenditure (₹) | 100 – 150 | 150 – 200 | 200 – 250 | 250 – 300 | 300 – 350 |
|---|---|---|---|---|---|
| Number of households | 4 | 5 | 12 | 2 | 2 |

SOLUTION

Use the step-deviation method with $a = 225$ and $h = 50$.

| $x_i$ | 125 | 175 | 225 | 275 | 325 | Total |
|---|---|---|---|---|---|---|
| $f_i$ | 4 | 5 | 12 | 2 | 2 | 25 |
| $u_i$ | $-2$ | $-1$ | 0 | 1 | 2 | |
| $f_i u_i$ | $-8$ | $-5$ | 0 | 2 | 4 | $-7$ |

$$\text{Mean} = 225 + \frac{-7}{25} \times 50 = 225 - 14 = 211.$$

**Answer: ₹211**

**Q7** The concentration of $SO_2$ in the air (in ppm) in 30 localities is given below. Find the mean concentration.

| Concentration (ppm) | 0.00 – 0.04 | 0.04 – 0.08 | 0.08 – 0.12 | 0.12 – 0.16 | 0.16 – 0.20 | 0.20 – 0.24 |
|---|---|---|---|---|---|---|
| Frequency | 4 | 9 | 9 | 2 | 4 | 2 |
| $x_i$ | 0.02 | 0.06 | 0.10 | 0.14 | 0.18 | 0.22 |
| $f_i x_i$ | 0.08 | 0.54 | 0.90 | 0.28 | 0.72 | 0.44 |

SOLUTION

$\sum f_i = 30$ and $\sum f_i x_i = 2.96$.

$$\text{Mean} = \frac{2.96}{30} \approx 0.099.$$

**Answer: about 0.099 ppm**

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**Q8** The absentee record of 40 students for a term is given below. Find the mean number of days a student was absent.

| Number of days | 0 – 6 | 6 – 10 | 10 – 14 | 14 – 20 | 20 – 28 | 28 – 38 | 38 – 40 |
|---|---|---|---|---|---|---|---|
| Number of students | 11 | 10 | 7 | 4 | 4 | 3 | 1 |

SOLUTION

The class widths are unequal, so use the direct method.

| $x_i$ | 3 | 8 | 12 | 17 | 24 | 33 | 39 | Total |
|---|---|---|---|---|---|---|---|---|
| $f_i x_i$ | 33 | 80 | 84 | 68 | 96 | 99 | 39 | 499 |

Mean $= \dfrac{499}{40} = 12.475 \approx 12.48.$

**Answer: about 12.48 days**

**Q9** The literacy rates (in %) of 35 cities are given below. Find the mean literacy rate.

| Literacy rate (%) | 45 – 55 | 55 – 65 | 65 – 75 | 75 – 85 | 85 – 95 |
|---|---|---|---|---|---|
| Number of cities | 3 | 10 | 11 | 8 | 3 |

SOLUTION

Use the step-deviation method with $a = 70$ and $h = 10$.

| $x_i$ | 50 | 60 | 70 | 80 | 90 | Total |
|---|---|---|---|---|---|---|
| $u_i$ | $-2$ | $-1$ | 0 | 1 | 2 | |
| $f_i u_i$ | $-6$ | $-10$ | 0 | 8 | 6 | $-2$ |

Mean $= 70 + \dfrac{-2}{35} \times 10 = 70 - 0.57 = 69.43.$

**Answer: about 69.43%**

## 2 Exercise 13.2

**Q1** The ages of patients admitted to a hospital in a year are given below. Find the mode and the mean, and compare them.

| Age (years) | 5 – 15 | 15 – 25 | 25 – 35 | 35 – 45 | 45 – 55 | 55 – 65 |
|---|---|---|---|---|---|---|
| Number of patients | 6 | 11 | 21 | 23 | 14 | 5 |

SOLUTION

The modal class is 35 – 45, with $l = 35$, $h = 10$, $f_1 = 23$, $f_0 = 21$, $f_2 = 14$.

Mode $= l + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 35 + \dfrac{2}{46 - 35} \times 10 = 35 + 1.82 = 36.8$ years.

For the mean, the class marks are 10, 20, 30, 40, 50, 60: $\sum f_i x_i = 60 + 220 + 630 + 920 + 700 + 300 = 2830$ and $\sum f_i = 80$. Mean $= \dfrac{2830}{80} = 35.37$ years.

The largest number of patients are about 36.8 years old, while the average age of a patient is about 35.37 years.

**Answer: mode $\approx 36.8$ years, mean $\approx 35.37$ years**

**Q2** The lifetimes (in hours) of 225 electrical components are given below. Find the modal lifetime.

| Lifetime (hours) | 0 – 20 | 20 – 40 | 40 – 60 | 60 – 80 | 80 – 100 | 100 – 120 |
|---|---|---|---|---|---|---|
| Frequency | 10 | 35 | 52 | 61 | 38 | 29 |

SOLUTION

The modal class is 60 – 80, with $l = 60$, $h = 20$, $f_1 = 61$, $f_0 = 52$, $f_2 = 38$.

Mode $= 60 + \dfrac{61 - 52}{122 - 52 - 38} \times 20 = 60 + \dfrac{9}{32} \times 20 = 65.625$.

**Answer: 65.625 hours**

**Q3** The monthly household expenditure of 200 families is given below. Find the modal and the mean monthly expenditure.

| Expenditure (₹) | 1000 – 1500 | 1500 – 2000 | 2000 – 2500 | 2500 – 3000 | 3000 – 3500 | 3500 – 4000 | 4000 – 4500 | 4500 – 5000 |
|---|---|---|---|---|---|---|---|---|
| Number of families | 24 | 40 | 33 | 28 | 30 | 22 | 16 | 7 |

SOLUTION

The modal class is 1500 – 2000, with $l = 1500$, $h = 500$, $f_1 = 40$, $f_0 = 24$, $f_2 = 33$.

Mode $= 1500 + \dfrac{40 - 24}{80 - 24 - 33} \times 500 = 1500 + \dfrac{16}{23} \times 500 \approx 1847.83.$

For the mean use $a = 2750$, $h = 500$. The values of $u_i$ are $-3, -2, -1, 0, 1, 2, 3, 4$, so $\sum f_i u_i = -72 - 80 - 33 + 0 + 30 + 44 + 48 + 28 = -35.$

Mean $= 2750 + \dfrac{-35}{200} \times 500 = 2750 - 87.5 = 2662.5.$

**Answer: mode ≈ ₹1847.83, mean = ₹2662.50**

**Q4** The state-wise teacher-student ratio in higher secondary schools is given below. Find the mode and the mean, and interpret them.

| Students per teacher | 15 – 20 | 20 – 25 | 25 – 30 | 30 – 35 | 35 – 40 | 40 – 45 | 45 – 50 | 50 – 55 |
|---|---|---|---|---|---|---|---|---|
| Number of states / U.T. | 3 | 8 | 9 | 10 | 3 | 0 | 0 | 2 |

SOLUTION

The modal class is 30 – 35, with $l = 30$, $h = 5$, $f_1 = 10$, $f_0 = 9$, $f_2 = 3$.

Mode $= 30 + \dfrac{10 - 9}{20 - 9 - 3} \times 5 = 30 + 0.625 \approx 30.6.$

For the mean, the class marks are 17.5, 22.5, ..., 52.5: $\sum f_i x_i = 52.5 + 180 + 247.5 + 325 + 112.5 + 0 + 0 + 105 = 1022.5$ and $\sum f_i = 35$. Mean $= \dfrac{1022.5}{35} \approx 29.2.$

Most states and U.T.s have about 30.6 students per teacher, and on average there are about 29.2 students per teacher.

**Answer: mode ≈ 30.6, mean ≈ 29.2**

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**Q5** The runs scored by some top batsmen in one-day internationals are given below. Find the mode.

| Runs scored | 3000 – 4000 | 4000 – 5000 | 5000 – 6000 | 6000 – 7000 | 7000 – 8000 | 8000 – 9000 | 9000 – 10000 | 10000 – 11000 |
|---|---|---|---|---|---|---|---|---|
| Number of batsmen | 4 | 18 | 9 | 7 | 6 | 3 | 1 | 1 |

SOLUTION

The modal class is 4000 – 5000, with $l = 4000$, $h = 1000$, $f_1 = 18$, $f_0 = 4$, $f_2 = 9$.

$$\text{Mode} = 4000 + \frac{18 - 4}{36 - 4 - 9} \times 1000 = 4000 + \frac{14}{23} \times 1000 \approx 4608.7.$$

**Answer: about 4608.7 runs**

**Q6** The number of cars passing a spot was noted for 100 periods of 3 minutes each, as below. Find the mode.

| Number of cars | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 | 40 – 50 | 50 – 60 | 60 – 70 | 70 – 80 |
|---|---|---|---|---|---|---|---|---|
| Frequency | 7 | 14 | 13 | 12 | 20 | 11 | 15 | 8 |

SOLUTION

The modal class is 40 – 50, with $l = 40$, $h = 10$, $f_1 = 20$, $f_0 = 12$, $f_2 = 11$.

$$\text{Mode} = 40 + \frac{20 - 12}{40 - 12 - 11} \times 10 = 40 + \frac{80}{17} \approx 44.7.$$

**Answer: about 44.7 cars**

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## 3 Exercise 13.3

**Q1** The monthly electricity consumption of 68 consumers is given below. Find the median, mean and mode, and compare them.

| Consumption (units) | 65 – 85 | 85 – 105 | 105 – 125 | 125 – 145 | 145 – 165 | 165 – 185 | 185 – 205 |
|---|---|---|---|---|---|---|---|
| Number of consumers | 4 | 5 | 13 | 20 | 14 | 8 | 4 |
| Cumulative frequency | 4 | 9 | 22 | 42 | 56 | 64 | 68 |

SOLUTION

Median: $\frac{n}{2} = 34$, so the median class is 125 – 145, with $l = 125$, $cf = 22$, $f = 20$, $h = 20$. Median $= l + \frac{\frac{n}{2} - cf}{f} \times h = 125 + \frac{34 - 22}{20} \times 20 = 137$.

Mode: the modal class is 125 – 145, with $f_1 = 20$, $f_0 = 13$, $f_2 = 14$. Mode $= 125 + \frac{7}{40 - 27} \times 20 \approx 135.76$.

Mean: with $a = 135$, $h = 20$ and $u_i = -3, -2, -1, 0, 1, 2, 3$, $\sum f_i u_i = -12 - 10 - 13 + 0 + 14 + 16 + 12 = 7$. Mean $= 135 + \frac{7}{68} \times 20 \approx 137.06$.

The three measures are close to one another, at about 137 units.

**Answer: median 137 units, mean $\approx 137.06$ units, mode $\approx 135.76$ units**

**Q2** The median of the distribution below is 28.5 and the total frequency is 60. Find $x$ and $y$.

| Class interval | 0 – 10 | 10 – 20 | 20 – 30 | 30 – 40 | 40 – 50 | 50 – 60 |
|---|---|---|---|---|---|---|
| Frequency | 5 | $x$ | 20 | 15 | $y$ | 5 |

SOLUTION

$5 + x + 20 + 15 + y + 5 = 60$ gives $x + y = 15$.

The median 28.5 lies in 20 – 30, so $l = 20$, $f = 20$, $h = 10$, $cf = 5 + x$ and $\frac{n}{2} = 30$.

$28.5 = 20 + \frac{30 - (5 + x)}{20} \times 10$ gives $8.5 = \frac{25 - x}{2}$, so $x = 8$ and $y = 7$.

**Answer:** $x = 8$, $y = 7$

**Q3** The ages of 100 policy holders are given as a "less than" distribution: below 20: 2, below 25: 6, below 30: 24, below 35: 45, below 40: 78, below 45: 89, below 50: 92, below 55: 98, below 60: 100. Policies are given only to persons aged 18 or more but less than 60. Find the median age.

SOLUTION

Subtract each cumulative frequency from the next to get the class frequencies.

| Age (years) | 18 – 20 | 20 – 25 | 25 – 30 | 30 – 35 | 35 – 40 | 40 – 45 | 45 – 50 | 50 – 55 | 55 – 60 |
|---|---|---|---|---|---|---|---|---|---|
| Frequency | 2 | 4 | 18 | 21 | 33 | 11 | 3 | 6 | 2 |
| Cumulative frequency | 2 | 6 | 24 | 45 | 78 | 89 | 92 | 98 | 100 |

$\frac{n}{2} = 50$, so the median class is 35 – 40, with $l = 35$, $cf = 45$, $f = 33$, $h = 5$.

Median $= 35 + \frac{50 - 45}{33} \times 5 \approx 35.76$.

**Answer: about 35.76 years**

**Q4** The lengths of 40 leaves, measured to the nearest millimetre, are given below. Find the median length.

| Length (mm) | 118 – 126 | 127 – 135 | 136 – 144 | 145 – 153 | 154 – 162 | 163 – 171 | 172 – 180 |
|---|---|---|---|---|---|---|---|
| Number of leaves | 3 | 5 | 9 | 12 | 5 | 4 | 2 |
| Cumulative frequency | 3 | 8 | 17 | 29 | 34 | 38 | 40 |

SOLUTION

The classes are not continuous, so change them to 117.5 – 126.5, 126.5 – 135.5, ..., 171.5 – 180.5.

$\frac{n}{2} = 20$, so the median class is 144.5 – 153.5, with $l = 144.5$, $cf = 17$, $f = 12$, $h = 9$.

Median $= 144.5 + \frac{20 - 17}{12} \times 9 = 144.5 + 2.25 = 146.75$.

**Answer: 146.75 mm**

**Q5** The lifetimes of 400 neon lamps are given below. Find the median lifetime.

| Lifetime (hours) | 1500 – 2000 | 2000 – 2500 | 2500 – 3000 | 3000 – 3500 | 3500 – 4000 | 4000 – 4500 | 4500 – 5000 |
|---|---|---|---|---|---|---|---|
| Number of lamps | 14 | 56 | 60 | 86 | 74 | 62 | 48 |
| Cumulative frequency | 14 | 70 | 130 | 216 | 290 | 352 | 400 |

SOLUTION

$\frac{n}{2} = 200$, so the median class is 3000 – 3500, with $l = 3000$, $cf = 130$, $f = 86$, $h = 500$.

Median $= 3000 + \frac{200 - 130}{86} \times 500 \approx 3406.98$.

**Answer: about 3406.98 hours**

**Q6** The numbers of letters in 100 surnames are given below. Find the median, the mean and the modal size.

| Number of letters | 1 – 4 | 4 – 7 | 7 – 10 | 10 – 13 | 13 – 16 | 16 – 19 |
|---|---|---|---|---|---|---|
| Number of surnames | 6 | 30 | 40 | 16 | 4 | 4 |
| Cumulative frequency | 6 | 36 | 76 | 92 | 96 | 100 |

SOLUTION

Median: $\frac{n}{2} = 50$, so the median class is 7 – 10, with $l = 7$, $cf = 36$, $f = 40$, $h = 3$. Median $= 7 + \frac{50 - 36}{40} \times 3 = 8.05$.

Mean: the class marks are 2.5, 5.5, 8.5, 11.5, 14.5, 17.5, so $\sum f_i x_i = 15 + 165 + 340 + 184 + 58 + 70 = 832$. Mean $= \frac{832}{100} = 8.32$.

Mode: the modal class is 7 – 10, with $f_1 = 40$, $f_0 = 30$, $f_2 = 16$. Mode $= 7 + \frac{10}{80 - 46} \times 3 \approx 7.88$.

**Answer: median 8.05, mean 8.32, mode $\approx 7.88$ letters**

**Q7** The weights of 30 students are given below. Find the median weight.

| Weight (kg) | 40 – 45 | 45 – 50 | 50 – 55 | 55 – 60 | 60 – 65 | 65 – 70 | 70 – 75 |
|---|---|---|---|---|---|---|---|
| Number of students | 2 | 3 | 8 | 6 | 6 | 3 | 2 |
| Cumulative frequency | 2 | 5 | 13 | 19 | 25 | 28 | 30 |

SOLUTION

$\frac{n}{2} = 15$, so the median class is 55 – 60, with $l = 55$, $cf = 13$, $f = 6$, $h = 5$.

Median $= 55 + \frac{15 - 13}{6} \times 5 \approx 56.67$.

**Answer: about 56.67 kg**

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